

PHYSICAL ACTIVITY & BMI PERCENTILE

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RESEARCH QUESTION

Do students who meet the recommended activity level (≥ 5 days/week, ≥ 60 min/day) have different BMI percentiles?

METHOD

One-Way ANOVA

$\alpha = 0.05$ · $n = 12,894$

VARIABLES

Categorical

Continuous

PhysicalActivity50rMoreDays

BMIPCT (BMI Percentile)

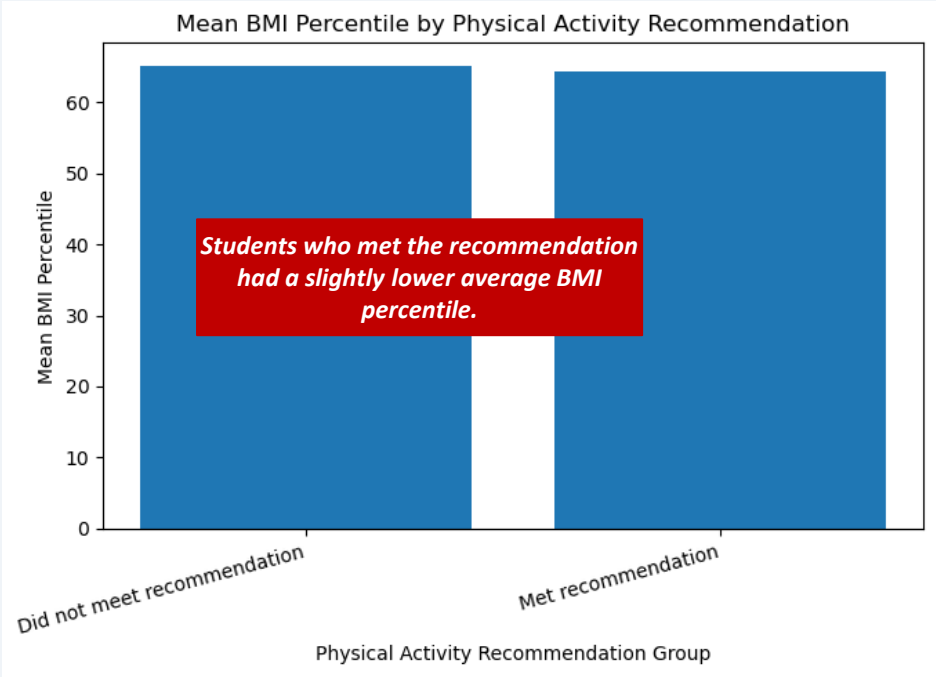
Limitations

- Self-reported data ·
- BMI influenced by diet & genetics ·
- Observational design
- 2007 data may not represent current adolescents

AN ANOVA ANALYSIS

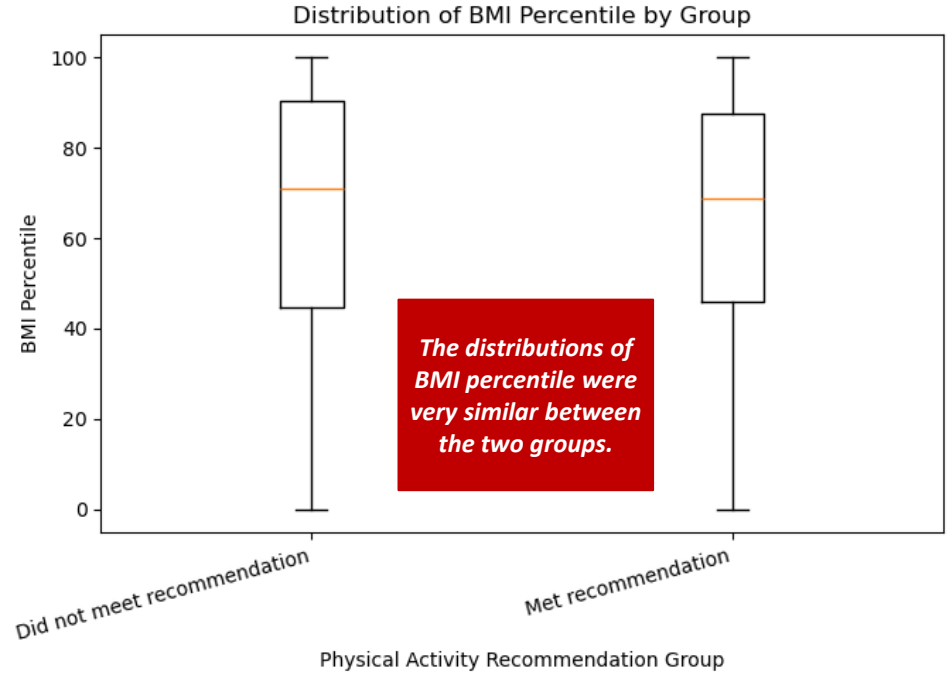
BAR CHART

Mean BMI Percentile by Physical Activity Group



DISTRIBUTION (BOXPLOT)

BMI Percentile by Group



64.38 vs 65.10

Mean BMIPCT
(Met vs Did Not Meet)

-0.71

pts difference

1.943

F statistic

0.163

p-value

[-1.70, 0.27]

95% CI

CONCLUSION We failed to reject the null hypothesis ($p = 0.163$). Insufficient evidence to conclude that students who met the physical activity recommendation had different mean BMI percentiles than those who did not.